Hoomwork

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Remember that you need to answer the questions but more importantly, you need to explain your thought process through the questions.

Question 1

Please encode the following polynomials in a form that is easily processed by a computer. It is not be *necessary* to use MATLAB code for this.

i. $x^2 + x^1 + 0$

ii. $y^2 + 5y^1$

iii. $25z^10 + 2z^5 + 20$

iv. $c_0x^4 + c_1x^3 + c_2x^2, c_3x^1, c_4x^0$

Question 2

Please decode the following coefficient row vectors to their respective polynomials. These polynomials should be simplified, thus no including those *parts* that are zero.

v. $\vec{c} = [4, 3, 2, 1]$

vi. $\vec{a} = [25, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1]$

vii. $\vec{v} = [1]$

viii. $\vec{r} = [4, 5, 6, 0, 0, 0, 1, 0, 2, 0, 0, 0, 1]$

Question 3

Answer the following to the best of your ability, These are all questions about polynomial encoding.

1. Does the variable or *part* name matter?
2. How is the degree of the polynomial encoded?
3. Why is it helpful to encode the polynomials? Elaborate.

Question 4

Please show in MATLAB code how you would encode and plot the following polynomial. $25x^5 + 10x^2 + 5$

Question 5

For the following situation is it appropriate to use interpolation or regression?

You are given data that has a large whole in the middle and have been asked to estimate that data.

Question 6

For the following situation is it appropriate to use interpolation or regression?

You have been given data that is extensive in several dimensions and you have been asked to estimate more data from all existing data on a single dimension.